

Table 1: A few categories for experimental comparisons. Full list in Appendix B.

Category	Description
Symbolic and algorithmic	Tasks involving symbol manipulation which can be solved by executing a program. This includes entity tracking datasets (e.g., SCONE, Coin Flip) and algorithmic tasks (e.g., BBH word sorting or finding shortest paths in a graph).
Math	Tasks requiring mathematical reasoning, from grade-school math to advanced mathematics, including physics questions.
Logical reasoning	Tasks designed to test for logical reasoning, whether deductive (Saparov & He, 2023, PrOntoQA), inductive (Bowen et al., 2024) or analogical (Ma et al., 2024) reasoning, including syllogisms and logical puzzles.
Encyclopedic knowledge	Tasks requiring expert-level in-depth knowledge beyond mere common-sense, usually in an open-book setting.
Mixed datasets	Datasets containing a variety of tasks, such as BIG-Bench Hard (BBH) or MMLU.
...	...

3 RESULTS FROM THE LITERATURE

Criteria and Process We investigate all papers from ICLR 2024, a representative ML venue, and two representative NLP venues, EACL 2024 and NAACL 2024 (including Findings and Workshop papers). This resulted in 4,642 papers total that filtered using automatic and manual methods to papers including experiments comparing chain-of-thought, $\mathcal{I}_{\text{cot}}$, against direct answering prompts, $\mathcal{I}_{\text{direct}}$. A total of 110 papers were found that matched our criteria with 1,218 experimental comparisons. We then grouped the comparisons by the types of tasks and datasets being evaluated. More details on our automatic and manual filtering, as well as our categorization, can be found in Appendix A and B.

Results Figure 2 shows the distribution of CoT deltas (CoT prompt minus the direct answer prompt performance) across our categorization of different task types found in the literature. Compared to Figure 1, we take the mean results per paper per category, indicated by blue dots, showing the trend across papers in the literature. The categories are ranked in order of ascending median CoT delta. The three categories which benefited the most from CoT are symbolic reasoning, math, and logical reasoning, with average improvements of 14.2, 12.3, 6.9, respectively. Average performance on these top three tasks with CoT was 56.9, whereas performance without CoT was 45.5. For other categories, the average performance with CoT was 56.8, compared to 56.1 without CoT. We do not consider this small improvement a victory for CoT. CoT involves more computation than direct answering, and a truly fair comparison between the methods should match the compute of the two methods, e.g., ensembling across multiple prompts.

Do any non-math datasets benefit from CoT? On the right side of Figure 2, we show the top 10 outliers from our observed trend, namely papers with high CoT deltas averaged across experiments in tasks *other than* math, symbolic, or logical reasoning. Although not categorized as math or logic, several of these are related to logical, mathematical or symbolic reasoning in some way. From this list, the dataset which benefits the most from CoT is BIG-bench Hard (BBH) (Suzgun et al., 2023), a benchmark consisting largely of problems requiring algorithmic, arithmetic or logical reasoning. For instance, BIG-bench Navigate is a spatial reasoning task, but relies heavily on a mathematical primitive of counting steps taken to derive a final conclusion. Similarly, while BIG-bench Temporal is a temporal reasoning task (answering questions about when certain events could have occurred), it requires deductive reasoning to solve. In addition, Legal Argument Reasoning (SemEval-2024 Task 5) (Bongard et al., 2022) was categorized as *context-aware QA*, but also requires substantial reasoning ability. Finally, MMLU-Moral Scenarios (Hendrycks et al., 2021a) requires answering two independent questions at once, which essentially involves a symbolic combination of two simpler questions.

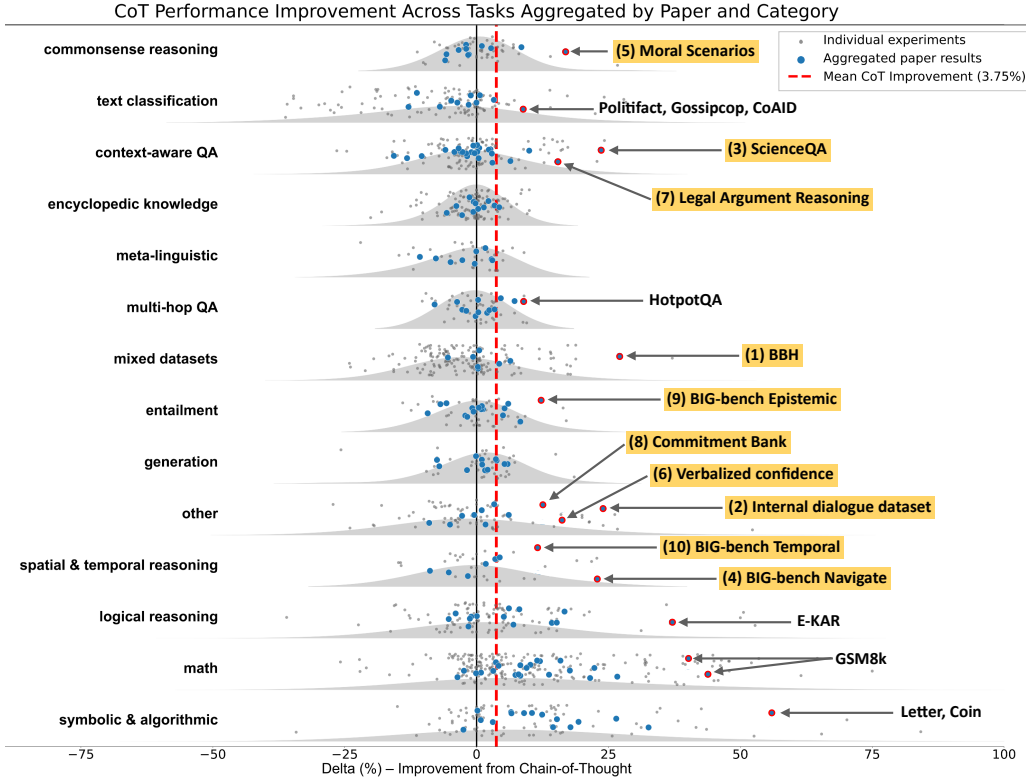


Figure 2: Results from our meta-analysis (grey dots) aggregated by paper and category (blue dots).

There are a few outliers that less clearly follow the trend. ScienceQA (Lu et al., 2022) consists of multiple choice questions across a range of natural and social science disciplines, though it is hard to interpret gains without knowing breaking down performance by subject or question type. The dialogue evaluation dataset from Jia et al. (2024) sees large improvements with CoT, but this is a proprietary dataset, and we note that other essay scoring results in our meta-analysis (Li et al., 2024; Stahl et al., 2024) did not show improvements with CoT. Other non-math, symbolic or logical datasets that benefit from CoT are Commitment Bank (de Marneffe et al., 2019) and the task of eliciting verbalized confidence (Xiong et al., 2024). Nevertheless, these are exceptions to the rule. The majority of the reported benefits from using CoT in the NLP and ML literature comes from math or math-related tasks.

4 RESULTS FROM EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Dataset, Models, Prompts All datasets, models, and prompts we evaluate over can be found in detail in the tables 3, 4, and 5 of Appendix C. We restricted our experiments to English models commonly used and benchmarked on general reasoning datasets. Our datasets include those which are widely used in CoT and reasoning literature, including a mix of non-symbolic, semisymbolic, and symbolic reasoning. They span different formats, including multiple-choice, short-answer, and free-response; however, most of these datasets are multiple choice or short answer, as CoT is not typically used in long-form response settings. We also categorize each dataset into a larger category of reasoning required to solve it: Commonsense, Knowledge, Symbolic, Mathematical, and Soft Reasoning. We define Soft Reasoning as questions relying on commonsense and natural language but going beyond simple inferences about these statements. Finally, we explore several prompting strategies for eliciting reasoning from language models, as past work has emphasized the importance of the prompt (Yang et al., 2024). However, we generally found slight performance differences; see Appendix D for details. We therefore focus on prompts similar to Kojima et al. (2022) and Wei et al.

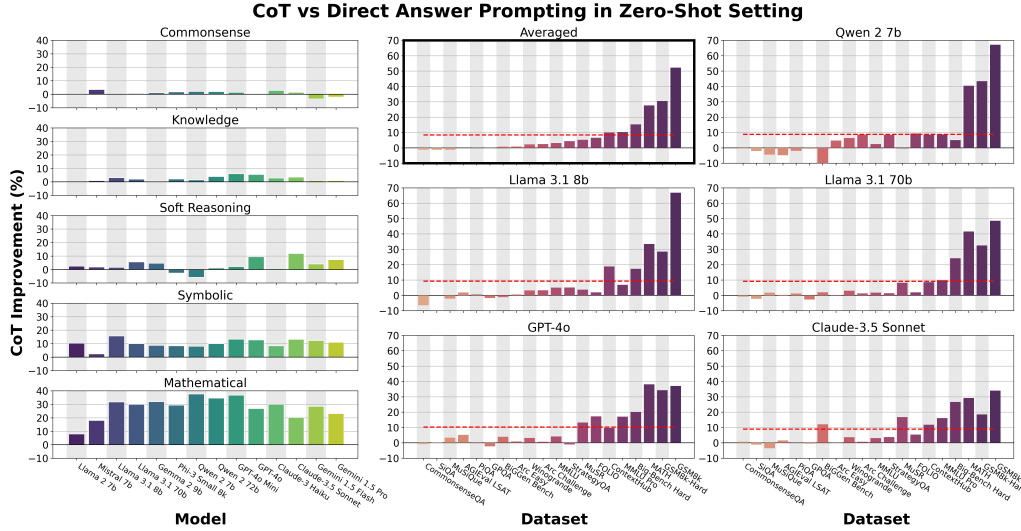


Figure 3: Left: Performance gain from using CoT for each reasoning category. Right: Performance gain from using CoT for each dataset, averaged across models and broken out across 5 representative models. Red lines indicate median improvement. In both plots we see a consistent trend: most improvements from using CoT are from math and symbolic reasoning.

(2022) for zero-shot and few-shot settings, respectively, with alterations to improve the model’s ability to produce desired behavior (i.e., formats that allow for easily parsed answers). We upload all our prompts and outputs for each model for each prompting strategy on Huggingface.⁴

Implementation Details We use a high-throughput inference package, vLLM (Kwon et al., 2023), for the model inference process. We use greedy decoding on all models. Our prompts are taken from the Llama 3.1 evaluations when available (Dubey et al., 2024), and minor adjustments are made to unify prompting strategies. For other datasets, we either use the standard prompt for the dataset from the corresponding original paper or implement our own prompt. Our answer parser (extract) is tailored to each dataset and model. Specific details about each dataset, its prompts, and answer extractor can be found in Appendix C.

4.2 RESULTS

Where does zero-shot CoT improve over direct prompts? *On datasets that require math (MATH, GSM8K) or formal logic (ContextHub, MuSR to a lesser degree) to answer the problem.*

Figure 3 on the left shows the average CoT performance improvement for each reasoning category from Figure 1 (right); raw numbers can be found in Table 6 of the Appendix. On the right, Figure 3 shows the performance gain from using CoT for each dataset, averaged across all models and for a selection of individual models. On non-symbolic reasoning categories and datasets, specifically those that contain questions primarily involving commonsense (CSQA, PIQA, SiQA), language understanding (WinoGrande), and reading comprehension (AGI LSAT, ARC-Easy, ARC-Challenge), there is little to no separation between the performance of zero-shot CoT and zero-shot direct answer. Despite these datasets involving reasoning, CoT does not yield improvement.

By contrast, the mathematical and symbolic categories get larger boosts in improvements alongside symbolic and many semi-symbolic datasets. MATH and GSM8K show gains as large as 41.6% and 66.9%, respectively. The semi-symbolic datasets like ContextHub and MuSR Murder Mysteries show moderate gains. These datasets require the application of logical rules to reach the answer, e.g., first-order logic parsed from simple natural language (ContextHub) or more complex commonsense statements (MuSR Murder Mysteries). All results are shown in the Appendix F.1 as well as a full list of numeric results for both CoT and direct answer prompting in Table 7. We also explored the few-shot setting and found it had little impact on when CoT will help; see Appendix E.

⁴<https://huggingface.co/collections/TAUR-Lab/cot-analysis-project-66bbb9e5e0156e65059895f5>